

Edge Hill University

The Department of Computer Science

Using pose estimation to improve amateur athletes' performance.

CIS3140 – Project report

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Abstract

In the field of data-driven sports analytics, the use of pose estimation has been used to gather information on sportsperson posture data and provide a posture comparison to a higher-level sportsperson, returning data on the difference in posture. However, often this data is not developed so that its readability is quality enough that it could be employed in a real-world scenario. This system aims to use golf swing segmentation techniques and pose analysis techniques to return, high-quality feedback through a keypoint comparison of an amateur golfer to a professional golfer. With this proposed system, amateur golfers should be able to improve their golf swing.

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Chapter 1: Introduction

Background and motivation

This project is being undertaken to find whether an amateur golf player's performance can be enhanced using machine learning and pose estimation. These concepts will be employed to provide insight into the errors the amateur players make in comparison to professional players and return feedback on the differences in technique. This system hypothesises that sports performance can be improved using these techniques and noticeable improvements in technique should be achievable.

While there are many popular research papers discussing pose estimation in sports, many of these papers only discuss the accuracy of their systems. Real-world application is one of the main points aims of this project, this is to assess the effectiveness of the system amongst amateur sportspeople.

The following papers offer no insight into real-world applications:

- Human Body Parts Estimation and Detection for Physical Sports Movements (JALAL, et al., 2019)
  - *This paper proposes a method of detecting robust human silhouette via salient regions and skin tone segmentation methods. –* ***93 citations***
- Player pose analysis in tennis video based on pose estimation (KUROSE, et al., 2018)
  - *This paper uses pose estimation to predict the probability of tennis play success –* ***10 citations****.*
- Head and Upper Body Pose Estimation in Team Sport Videos (HAYASHI, et al., 2013)
  - *This paper proposes a head and upper body pose estimation framework for team sports videos with helmet players –* ***15 citations****.*

These are 3 of the more popular sport-related pose estimation papers on IEEE Xplore, Considering the lack of real-world application in this field, this project seeks to apply its golf swing feedback system, built through the use of a golf swing sequencing and segmentation system, and a pose estimation/analysis system, to a real-world context to explore the possibility of providing feedback that can improve the ability of sportspeople.

Aim and Objectives

The main aim of this project is to create a pose detection system that will attempt to enhance the sporting ability of an amateur golf player.

Objectives

- Retrieve professional and amateur golf footage for data pre-processing. This footage will ideally be a “down-the-line” camera angle in which the driver club is used.
- Build/use a deep learning system that can be used on golf footage to pre-process golf footage, the model should also be able to differentiate moments of the golf swing and store the image data of each moment of a golf swing in a corresponding file, and with a relevant filename for both the file directory and the file for data pre-processing.
- Build/use a pose estimation system, the system must return keypoint data of a reasonable quality that is readable and understandable. The system should return a high level of accuracy in its pose estimation to return accurate data.
- Using the image data from the deep learning model, apply the pose estimation system to the image data, returning keypoint data for each swing moment. This data will be written to a file with data for each keypoint stored.
- Compare the amateur and professional keypoint data and return meaningful feedback on the amateur's swing, assessing whether this feedback improves golf performance.

Scope / System proposal

The golf swing analysis system is built to improve amateur golfers through swing sequencing, segmentation of these sequences, and pose estimation. The sequencing system will identify 8 separate key moments of a swing, the segmentation system will return an image at each estimated moment. The pose estimation system will apply pose estimation techniques to these images and compare the keypoint data of the amateur golfer to the keypoint data of professional golfers.

To ensure that the image input data is of high quality, a rejection value will be implemented so that only videos that can be segmented into images that meet a specific accuracy level will be accepted for data analysis. This is to ensure that quality data is used throughout the system.

The system will use the 3D pose estimation library “Mediapipe” as its 3D pose estimation system allows for depth information to be captured. If it is found that this library does not provide ideal data, other pose estimation techniques, such as 2D pose estimation may be considered.

The Python programming language was chosen due to its strengths in lightweight development environments and data analytics (MCKINNEY, 2012) .

No SDLC is necessary due to the largely pre-built structure of this system, all data analysis programmes will be relatively small and therefore require no development lifecycle

To provide feedback on the golf swing, the amateur keypoint that is on average most distant to the professional average will be returned to the amateur golfer, providing inference on how they can improve the swing based on potential problems caused at this keypoint.

Overall, the scope of this system is to provide a data-driven approach to golf coaching and swing analysis, which returns clear feedback on how the golfer can improve their performance.

Methods

Data collection

Footage of an amateur golf player will be shot; various videos will be taken to find an average. Footage of professional golf players will be accessed through the GolfDB golf footage dataset.

Data cleansing

A CSV packaged with the GolfDB repository stores information about each swing video, with this, perform data cleansing on the video footage to assure only the desired footage is being tested. For this project, swings in which the club used is the “driver”, the view is "down-the-line" and slow = 0 are used. An example of this angle is shown in Fig 1. The GolfDB video repository will be cleaned to only store footage with these parameters.

Figure 1 - ideal camera angle



Data pre-processing

To pre-process this data, video footage of both the amateur and professional golf players will be processed through the GolfDB swing sequencing model. This model identifies 8 key moments of a golf swing, these being: Address, Toe-up, Mid-backswing, Top, Mid-downswing, Impact, Mid-follow-through, and Finish. To return these key moments for further data analysis, a segmentation system, packaged within the golfDB repository, will capture each key moment of the swing and each moment will be saved as an image to a file correlating with the moment of the swing for further testing.

Data implementation through pose estimation modules

The pre-processed image data returned by the footage segmentation system will be implemented into the pose estimation system through OpenCV. OpenCV is a computer vision and machine learning software library built to provide a common infrastructure for computer vision applications to accelerate the use of machine perception. OpenCV makes use of functions that allow for the reading and processing of image data to Python (OPENCV, 2023). The pose estimation system will likely be built with Mediapipe, which is a computer vision framework that supports 3D pose estimation (MEDIAPIPE, 2023). The pose estimation module of Mediapipe will be applied to the images saved from the image segmentation model. The keypoint data of the professional keypoints will be returned and written to a CSV file that will become the baseline data for keypoint data comparison.

Data evaluation

From the keypoint data returned, a comparison of the professional's swing to the amateur swing will be made. This will be through a comparison of keypoint locations between the professional and amateur player, where the average amateur keypoints that are furthest away from the professional average are returned as the 'area needing improvement'. As well as this, advice on how their swing can be bettered will be returned, and whether a performance improvement is made will be assessed.

Ethics

It will be especially important to assure that all the test subjects are willing to be recorded and have their results reported in an academic setting. Written consent may be required depending on the level at which the subjects are discussed in the paper. As well as this, the driving range must be aware of the recording and the plans to use the data gathered in academic research.

The test subjects must be comfortable throughout the recording as over-exertion is possible and is to be avoided at all costs. As well as this over-exertion can cause injuries and potentially muscle damage, to avoid any level of over-exertion the videos will be taken at a leisurely pace with no rush or pressure to perform to any standard.

No personal details about the test subject shall be used in the research unless permitted and necessary. For example, the height of the golfer may affect the pose estimation and would therefore need to be discussed, however only with the subject's permission.

Report structure

The report will consist of 6 chapters, these being:

- Chapter 1: introduction, an overview of the problem at hand, factors to consider, and the problems raised, including a discussion on how the problems will be tackled.
- Chapter 2: Background and Literature Review/Related Work. an introduction to the main concepts of the project topic and why this research is being undertaken. As well as a literature review critically evaluating related work and discussions on similar systems are opened.
- Chapter 3: Problem analysis/requirements. description of the potential challenges when building this system as well as an analysis of the functional and non-functional requirements of this system
- Chapter 4: Data gathering and analysis. Discussions of data-gathering techniques and the efficacy of these techniques are assessed. The pre-built models such as the swing segmentation, pose estimation, and feedback systems are employed, and the data returns are analysed.
- Chapter 5: system testing and evaluation. The project is deployed to a real-world scenario, and the feedback generated is applied to an amateur golfer to find whether improvements to the golfer's swing have been made.
- Chapter 6: conclusion/ critical evaluation. A summary of the project is provided, as well as a critical evaluation, which discusses all aspects of the system and how well it performed.

Chapter 2: Background / lit review

This project was influenced largely due to the perceived lack of accessible sports performance systems currently available. Apps such as Golf Swing Analyzer ++ and V1 Golf, two of the most popular golf improvement apps on the app market do not offer automated analysis of golf swing in the fashion this system plans to.

Golf swing sequencing is a deep learning technique introduced to provide a benchmark database for golf analysis systems (MCNALLY, et al., 2019). This system allows for the segmentation of footage of a golf swing into 8 different key moments of a golf swing, these being: Address, Toe-up, Mid-backswing, Top, Mid-downswing, Impact, Mid-follow-through, and Finish. These are shown from left to right in Fig 2.



Figure 2 - golf swing sequence

The GolfDB swing sequencing system uses MobileNetV2, which is a CNN-based inverted residual structure and makes efficient use of lightweight depthwise convolutions (HOWARD, et al., 2017). The temporal context is necessary to detect swing events. To extract the temporal information, the sequence of feature vectors $f\ (f_1, f_2, \dots, f_T)$, where f_t is an element of $\mathbb{R}^{1280}$ is found through the global mean pooling on the final feature map of MobileNetV2. The sequence is then input for an N-layer bidirectional LTSM (HOCHREITER, 1997), which has H hidden units in each layer. For every frame t, The H dimensional output of the LTSM is input through a final fully connected layer, and the application of a SoftMax function is used to compute class probabilities e. Through this, the weights of the fully connected layer are shared across all frames.

Research into Golf swing sequencing (MCNALLY, et al., 2019), has opened the door for golf technique analysis systems. Action recognition systems such as this have already been used in various fields of sports to improve player performance in sports such as handball (HOST, et al., 2022). Application of similar systems into other areas of sport such as sports physiotherapy have shown promising potential through application into statistical analysis models (RAHMAD, et al., 2018).

Pose estimation is the process of using machine learning models to predict the pose of a human with either a video or an image by estimating the spatial locations of key body joints (keypoints) (TENSORFLOW, 2023). Pose estimation can therefore be used to estimate keypoint location and data.

Research that presents pose estimation's potential for sports coaching make use of pose analysis, where poor posture is classified and sportspeople are told when poor posture is performed. AI coaching systems such as these have returned positive sentiment amongst a wide range of sportspeople, from enthusiasts to professionals (WANG, et al., 2019).

Related work

Overview of existing systems

Many recent studies open a dialogue into swing improvement in golf. The use of both physical and non-physical devices have been used to make inferences on a sportspersons golf swing, for example, the use of optical tracking cameras have been used to assess swing kinematic and ground reaction force data, where a physical device attached to each joint vibrates when said joint performs poorly during a golf swing (WOZNIAK, et al., 2020). Optical tracking cameras can capture data with a high level of accuracy as well as provide real-time feedback, which at the date of research was state-of-the-art (2020) (WOZNIAK, et al., 2020). However, systems that make use of physical devices for tracking sports movement are not accessible for the average golfer, due to the complexity of employing an these devices.

As well as this, 3D pose estimation algorithms have been researched for posture analysis of a golf swing through single image analysis, where each key moment of swing must be performed individually by the golfer, and if a keypoint is erroneous, its location is highlighted (PARK, et al., 2017). A focus on posture can help users to identify current issues with their swing, as well as provide a level of accessibility as no external devices are required, however, the referenced system provides a limited scope of analysis. By performing the moment of swings individually, and not within a natural golf swing, the true posture of the sportsperson at each moment cannot be confirmed, it could be theorized that the sportsperson would be paying more attention to their form at each moment of swing than they would in a natural golf swing and therefore the quality of form would be incorrect.

Importance of personalized feedback

A comparison between an amateur and a professional can provide inference that can improve amateur performance. Research suggests that amateur performance can be improved through a comparison to professional players. Amateur performance in Tai-Chi, was shown to be improved when compared to a professional Tai-Chi martial artist, where a visualization of standard movement resulted in an improvement in performance (HAN, et al., 2017). As well as this, research into pose comparison of a student dancer and a teacher performing dance moves would return feedback in the form of accuracy of the student's movements when compared to the teacher (RAPTIS, et al., 2011), which also resulted in an improvement. Feedback based around Pose comparison can return targeted and personalized areas for improvement.

Advances in computer vision and machine learning

Research into computer vision and ML applications within sports have shown fruitful results, deep learning methods such as Recurrent Neural networks (RNNs) and Convolutional neural networks (CNNs) have become popular due to their high-quality performance in action recognition and image classification tasks when applied to complex data (GUANGLE, et al., 2019). CNNs have demonstrated high-quality performance in action recognition and image classification tasks (BHATT, et al., 2021), (KHAN, et al., 2020), allowing for application into Golf swing segmentation systems such as GolfDB or Golf swing pose estimation tools (LIAO, et al., 2022).

Recent computer vision systems have proposed hybrid architectures that make use of both CNNs and RNNs. Deep convolutional Recurrent neural networks (C-RNNs) have shown promising performances in tasks such as object classification and frame prediction (SINHA, et al., 2018) (WAN, et al., 2018).

Pose estimation is a subfield of computer vision that makes use of deep learning techniques to track human poses in image and video data for applications in fields such as robotics, sports analysis, and video surveillance. Various approaches have been used to build accurate pose estimation systems. For example, a pose estimation method that uses convolutional deep neural networks outperformed other methods such as Body Parts Dependent Joint Regressors (DANTONE, et al., 2014) and Clustered Pose and Nonlinear Appearance Models (JOHNSON & EVERINGHAM, 2010). when trained on the FLIC and Leeds sports datasets. A method for multi-person pose estimation using Part Affinity Fields achieved high accuracy rates on the COCO dataset. (CAO, et al., 2017). Recently, attempts in improving the accuracy of pre-existing models have been researched. Mask R-CNN is a model that builds on Faster R-CNN (REN, et al., 2015) and has returned high accuracy rates in object instance segmentation (HE, et al., 2017).

Chapter 3: Problem Analysis/Requirements

Problem analysis

This system addresses the problem of sports coaching through ML and data-driven assessment. Typically, golf analysis would come from an expert such as a coach, but the opinions provided by these experts can be subjective. This system attempts to use deep learning techniques to efficiently automate a golf swing analysis to provide unbiased, data-driven feedback through a comparison to a professional sportsperson's swing data.

One of the primary challenges in building this system is the segmentation of key moments in golf footage for later analysis by a pose estimation system. Key moments in a golf swing video will need to be identified, however, the accuracy of the segmentation may vary depending on the golfer and the quality of the footage. Factors such as camera angles and lighting could be problematic to the quality of data.

The preparation of the image database will be a key challenge for this system due to the importance of the images to the pose estimation system. video databases of both professional and amateur golf players will need to be segmented and stored in their respective areas. Both video databases will need to be large due to the high possibility of erroneous data, especially within the amateur footage as the footage will likely be shot at a lower quality.

The pose estimation system must accurately identify the key points of the images and calculate average keypoint data for the professional golfer's image data at each key moment. Without the accurate identification of keypoints, quality of data cannot be assured.

A final system must be created to compare the keypoint data of the individual amateur performance against the professional average. This system will find the keypoints of the amateur that are furthest away from the professional average and return this information as meaningful feedback to the amateur golfers. This feedback must be specific and realistic, giving the amateur golfer clear advice on how to improve their swing.

Overall, this system will address the difficulty of improving amateur golf performance through the use of deep learning techniques to automatically provide analysis of a golf swing and return informative feedback through a comparison to professional golf performance. This system faces various obstacles such as data selection, pose estimation, and feedback delivery. These challenges must be meticulously assessed to ensure the success of the system.

Requirement analysis

Functional requirements

- The system will be able to segment the input video into different images that represent each key moment of the golf swing and save each image to an adjacent file as well as name the image file suitably in relation to the moment of swing.
- The system will be able to reject footage that does not return decent quality accuracy rates throughout the entire swing.
- The system will be able to apply a pose estimation system to each image to extract and write the key point data to a file.
- The system will be able to compute the average keypoint data for the professional player's swings and write the resulting data to a file.
- The system shall be able to compare the amateur player's average keypoint data to the professional player's averages. It should be able to identify the keypoints that are furthest away from the professional average.
- The system will be able to return feedback to the amateur player detailing how they could better their swing through returning problem areas that need improvement.

Non-Functional requirements

- The system should be able to return images that are as close to the moment of swing as possible, allowing for better data analytics.
- The system should be able to detect the key moments of the golf swing to a high level of accuracy as well as return a high level of accuracy when applying pose estimation techniques to an image.
- The system should not produce many false positives or false negatives, it should be dependable and consistent in its results.
- The system should be able to work with both low-quality and high-quality video, giving access to as many users as possible.
- The system should have a level of scalability, thus allowing for increasingly larger amounts of video to be processed at the same time.

Chapter 4: Data gathering and analysis.

Data cleansing

This project makes use of the benchmark golf swing database 'GolfDB'. GolfDB consists of 1400+ golf swing videos, each labelled with event frames, bounding boxes, player name and sex, club type, and view type (MCNALLY, et al., 2019). The sportspeople within these videos are all professional players and therefore are used for the professional dataset. As this system attempts the improve the 'Drive' swing within golf, the dataset must be cleansed so that only videos with this golf club are used. As well as this, the camera view for the video must be a 'down-the-line' angle, which means that the camera is pointing down the target line. This angle is ideal as it shows the full body within the golf swing, an example of a down-the-line camera angle is shown in Fig 1.

To sort through the video data, a .pkl file given in the GolfDB GitHub repository is converted to CSV, this is to make it accessible by Microsoft Excel and therefore easily viewable. This is done through a basic Python script shown in Fig 3. This allows for the finding of videos that meet the parameters.

Figure 3 .pkl to .csv

```
import pickle as pkl
import pandas as pd
with open("GolfDB.pkl", "rb") as f:
    object = pkl.load(f)

df = pd.DataFrame(object)
df.to_csv(r'GolfDB.csv')
```

The GolfDB video repository came with both normal and slowed-down footage. In testing, the segmentation system returned lower accuracy rates on slowed videos compared to the normal-speed footage. Fig 4 and Fig 5 show the accuracy rates of footage, where Fig 4 is a video at normal playback speed and Fig 5 is the same video at slow speeds.

```
Preparing video: data/videos_160(1)\0.mp4
Using device: cuda
Loaded model weights
Testing...
Predicted event frames: [48 63 69 83 87 90 92 92]
Confidence: [0.159]
Confidence: [0.159, 0.69]
Confidence: [0.159, 0.69, 0.731]
Confidence: [0.159, 0.69, 0.731, 0.401]
Confidence: [0.159, 0.69, 0.731, 0.401, 0.463]
Confidence: [0.159, 0.69, 0.731, 0.401, 0.463, 0.359]
Confidence: [0.159, 0.69, 0.731, 0.401, 0.463, 0.359, 0.099]
Confidence: [0.159, 0.69, 0.731, 0.401, 0.463, 0.359, 0.099, 0.002]
```

Figure 4 - normal playback speed accuracy rates

```
Preparing video: data/videos_160(1)\1.mp4
Using device: cuda
Loaded model weights
Testing...
Predicted event frames: [ 48  96 109 188 194 205 214 274]
Confidence: [0.049]
Confidence: [0.049, 0.494]
Confidence: [0.049, 0.494, 0.292]
Confidence: [0.049, 0.494, 0.292, 0.222]
Confidence: [0.049, 0.494, 0.292, 0.222, 0.256]
Confidence: [0.049, 0.494, 0.292, 0.222, 0.256, 0.137]
Confidence: [0.049, 0.494, 0.292, 0.222, 0.256, 0.137, 0.069]
Confidence: [0.049, 0.494, 0.292, 0.222, 0.256, 0.137, 0.069, 0.225]
```

Figure 5 - slowed down accuracy rates.

The accuracy rates in the earlier moments are higher on the normal-speed videos, but the final moments tend to be higher on the slowed-down footage.

It is theorized that backswing events (Address, Toe-up, Mid-backswing, Top) have lower detection rates within slowed footage since a backswing is typically taken much slower than a downswing. This means that there are more frames in the backswing that are similar in appearance to the ground-truth event frames (MCNALLY, et al., 2019). Due to this, normal speed footage is used.

To extract the ideal footage from the dataset, The ID of footage where Club = "driver", view = "down-the-line", and slow = "0" are extracted from the dataset using the Excel filter function, as shown in Fig 7. The ids of these rows are gathered, and a Python script is used to delete all videos but these from the dataset. Code is shown in Fig 6.

```python
import csv
import os

# File names to check against
filename = "data/DownTheLineList.csv"

# List to store potential names
potential_names = []

# Read the CSV file
with open(filename, "r") as file:
    reader = csv.reader(file)
    for row in reader:
        potential_names.append(row[2])

# Directory to check for files
dir_path = "data/Fast_Drive"

# Loop through all the files in the directory
for file_name in os.listdir(dir_path):
    # Check if the file name is in the potential names list
    if file_name not in potential_names:
        # If the file name is in the list, delete the file
        file_path = os.path.join(dir_path, file_name)
        os.remove(file_path)
        print(f"Deleted file: {file_name}")
```

Figure 6 - file deleter

id	slow	club	view
0	0	driver	down-the-line
2	0	driver	down-the-line
4	0	driver	down-the-line
6	0	driver	down-the-line
12	0	driver	down-the-line
33	0	driver	down-the-line
37	0	driver	down-the-line
43	0	driver	down-the-line
45	0	driver	down-the-line
55	0	driver	down-the-line
62	0	driver	down-the-line
70	0	driver	down-the-line
73	0	driver	down-the-line
76	0	driver	down-the-line
88	0	driver	down-the-line
94	0	driver	down-the-line
106	0	driver	down-the-line
108	0	driver	down-the-line
112	0	driver	down-the-line
119	0	driver	down-the-line
125	0	driver	down-the-line
130	0	driver	down-the-line
132	0	driver	down-the-line
134	0	driver	down-the-line
146	0	driver	down-the-line
168	0	driver	down-the-line
178	0	driver	down-the-line
184	0	driver	down-the-line
185	0	driver	down-the-line
189	0	driver	down-the-line

Figure 7 - ID capture

Model testing

Training custom models

golfDB makes use of a PyTorch implementation of the MobileNetV2 baseline model, naming their model 'Swingnet'. This model was evaluated against multiple parameters to find the configuration that yields the best results. This is measured by an evaluation metric that is named: 'Percentage of Correct Events' or PCE for short. PCE is measured by a value of tolerance denoted as **δ.** A measure of tolerance within event detection is ideal due to the difficulty in returning a perfectly spotted event (SILVIO, et al., 2018). Tolerance is calculated from the number of frames correctly detected, set by default to 1 (MCNALLY, et al., 2019). Fig 8 is the formula for calculating tolerance per moment of swing.

$$\boldsymbol{\delta = \max(\left\lfloor \frac{n}{f} \right\rceil, 1)}$$

Figure 8 - tolerance equation

Where n is the no. of frames from Address to impact, f is the sampling frequency and $\lfloor x \rceil$ rounds x to the nearest integer.

When a model is evaluated, a random sample of 25% of the video database is tested, with each video returning a Boolean value for whether it has been accurately spotted or not. This data is returned in a list, as shown in Fig 9. The mean of all lists are found, and the average is calculated, this is returned as the Average PCE.

```
343 [0 0 1 0 1 1 1 0]
344 [0 0 1 1 1 1 1 0]
345 [0 0 1 1 1 1 1 1]
346 [0 1 1 0 1 1 1 1]
347 [0 1 0 1 1 1 1 0]
348 [1 0 1 1 1 1 1 0]
349 [0 0 1 1 1 1 0 0]
Average PCE: 0.6503571428571429
```

Figure 9 - average PCE

The GolfDB system trains its models using the pre-trained MobileNetv2 weights and various standard training configurations such as sequence length and batch size. Through this approach, the GolfDB system generates a custom MobileNetv2 model named Swingnet. These training configurations can be changed to accommodate lower-quality hardware by training a custom Swingnet model specifically trained for use on a lower-quality GPU.

The developers of the GolfDB utilized a single NVIDIA Titan XP GPU for their model training purposes. The specs of this GPU are listed in Fig 10:

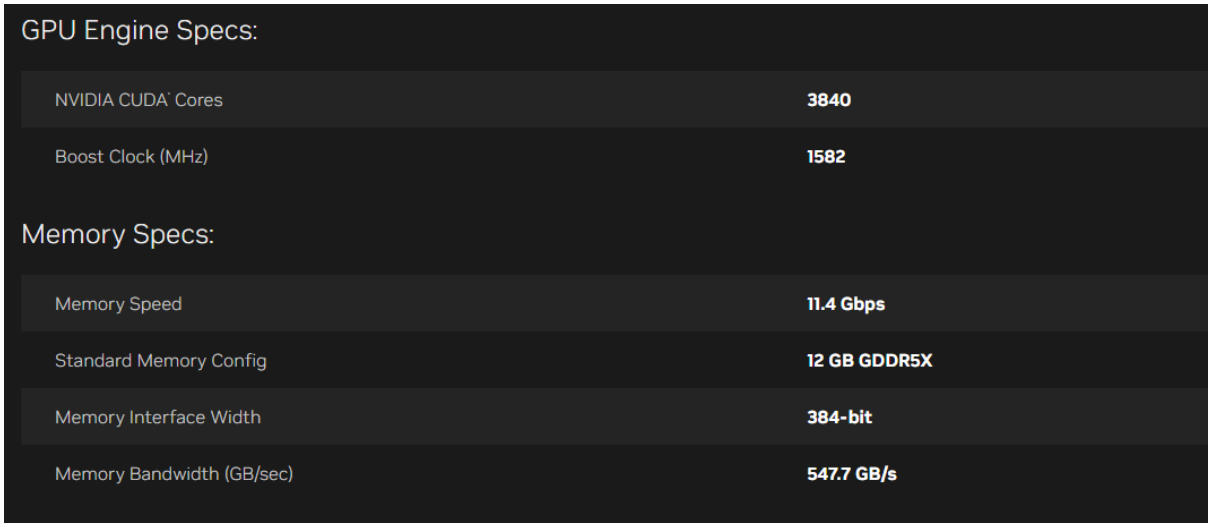


Figure 10 - Titan GPU Specs

The NVIDIA Titan XP GPU performs much better than the Nvidia 1060 6 GB, which is what was used throughout this project. The specs of the NVIDIA 1060 6gb are listed in Fig 11:

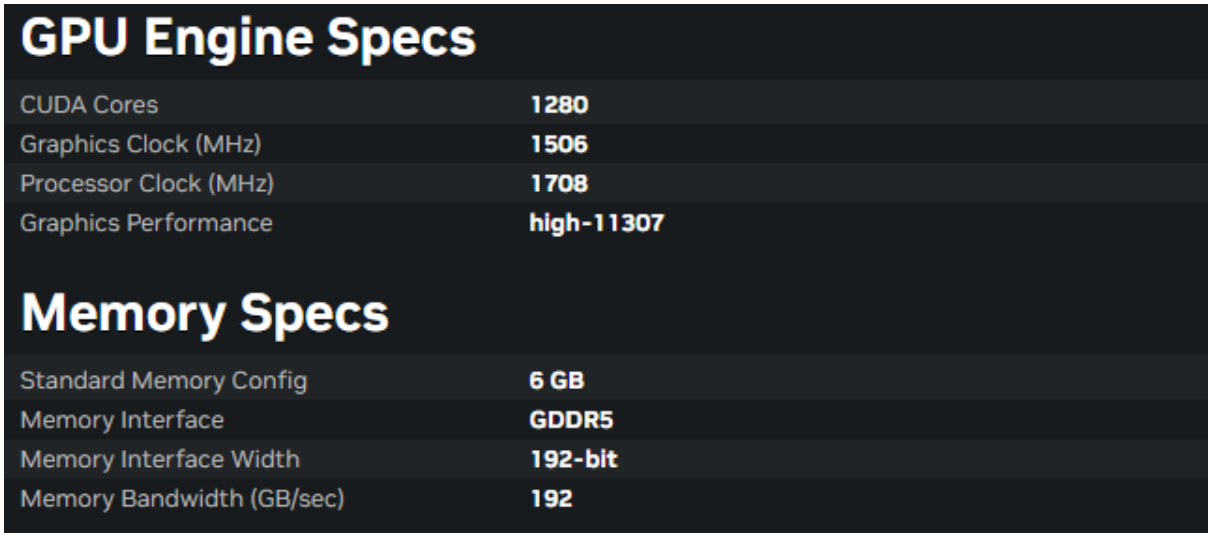


Figure 11 - Nvidia 1060 6gb specs

The GolfDB image segmentation system makes use of a PyTorch implementation of the Swingnet baseline model which employs CUDA cores to process logical operations within a GPU, therefore more Cores allow for faster computation time (NVIDIA, 2023). A higher amount of memory allows for more data to be processed by the GPU at one time. These factors allow for much faster training, and in circumstances such as this, a GPU of such lower quality cannot perform deep learning tasks to the same extent.

Due to the lower quality GPU, the batch size and sequence size had to be lowered during model training. This decrease in both the batch size and the sequence length drastically decreased the detection levels of the model.

This is because batch size represents the number of training samples that are used to make an update to the network parameters. This means it affects the amount of time to converge as well as the amount of overfitting. Utilizing a smaller batch size increases performance speed but may lower the detection rate due to fewer updates per training iteration (RADIUK, 2017).

Sequence Length represents input data length within RNNs, which represents the length of the input data for RNNs. Sequence length dictates the amount of information the RNN can access, and a lower sequence length can result in difficulties when attempting to learn temporal dependencies in the data (FANG, et al., 2021), thus resulting in reduced detection rates within the system.

Considering that swingnet is a deep hybrid convolution and recurrent network, both of these parameters are significant.

To train a swingnet model suitable for the hardware constraints, the batch size was reduced from 22 to 8, and the sequence length was reduced from 64 to 48. This resulted in a PCE loss of 0.074, where

the original model returns an average PCE of 0.715, and the custom model returns an average PCE of 0.641. This is shown in Fig 12 and Fig 13.

```
330 [0 0 1 1 1 1 1 0]
331 [0 1 1 1 0 0 0 0]
332 [0 0 1 1 1 1 1 0]
333 [0 0 1 1 1 1 0 0]
334 [0 1 1 1 1 1 1 0]
335 [0 1 1 1 1 1 1 0]
336 [0 1 1 1 1 1 1 0]
337 [0 1 1 0 1 1 1 0]
338 [0 1 1 1 1 1 1 0]
339 [0 0 1 0 1 1 1 0]
340 [0 0 1 0 1 1 1 0]
341 [0 0 0 1 1 1 1 0]
342 [0 1 1 1 1 1 1 0]
343 [0 1 1 0 1 1 1 0]
344 [0 0 1 1 1 1 1 1]
345 [0 0 1 0 1 1 1 1]
346 [0 1 1 1 1 1 1 0]
347 [0 1 1 1 1 1 1 0]
348 [0 0 1 1 1 1 1 0]
349 [0 0 1 0 1 1 1 1]
Average PCE: 0.6410714285714286
```

Figure 12 - custom model

```
330 [0 1 1 1 1 1 1 0]
331 [0 0 1 1 1 1 1 0]
332 [0 1 1 1 1 1 1 0]
333 [0 1 1 1 1 1 1 0]
334 [1 1 1 1 1 1 1 1]
335 [0 1 1 1 1 1 1 0]
336 [0 1 1 1 1 1 1 0]
337 [1 1 1 1 1 1 0 0]
338 [0 1 1 1 1 1 1 0]
339 [0 1 0 0 1 1 1 0]
340 [0 1 1 0 1 1 1 0]
341 [0 1 1 0 1 1 1 1]
342 [1 1 1 1 1 1 1 0]
343 [0 1 1 1 1 1 1 0]
344 [0 1 1 1 1 1 1 0]
345 [0 0 0 0 1 1 1 0]
346 [0 1 1 1 1 1 1 0]
347 [0 1 1 0 1 1 1 0]
348 [1 0 1 1 1 1 0 0]
349 [0 0 1 1 1 1 0 0]
Average PCE: 0.715
```

Figure 13 - pre-trained model

Considering the large drop-in detection rates, the use of a custom model was not ideal. Fortunately, access to the pre-trained Swingnet weights were made public by the GolfDB researchers, so the lower-quality model was not necessary. The model training was the only area that required a high-quality GPU, so the use of a lower-quality GPU in other areas of the system, such as the image segmentation system is of no concern.

Footage testing

The GolfDB swing sequencing model was trained on high quality footage, shot in ideal light and frame. When attempting to sequence and segment lower-quality footage, accuracy rates dropped, resulting in amateur video conditions becoming an issue. Footage that had poor lighting and framing returned noticeably low accuracy rates. Fig 14 and Fig 15 show two videos of the same test subject at the same moment of swing, where Fig 15 returned respectable detection rates and Fig 14 did not, as can be seen by the accuracy value in the top corner.

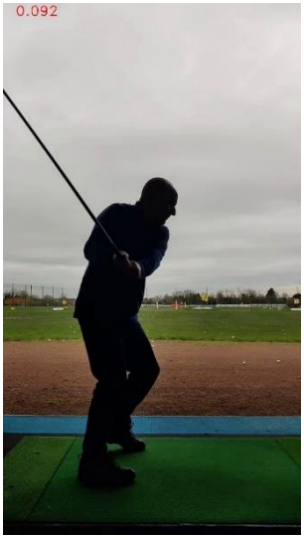


Figure 14 - poor lighting footage



Figure 15 - ideal lighting footage

By comparing the accuracy rates, it can be concluded that lighting is likely a key factor in detection accuracy, as Fig 14 returns an accuracy rate of 0.092 and Fig 15 returns an accuracy rate of 0.722, Where Fig 15 decidedly has better lighting.

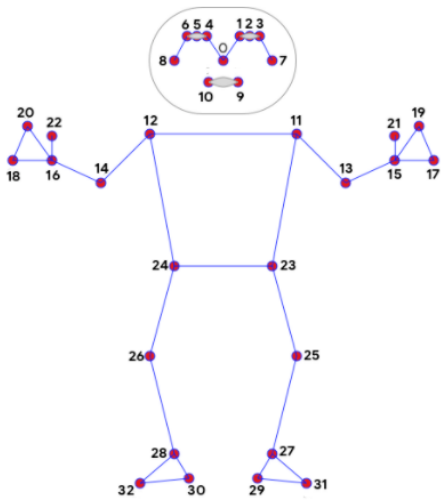
Pose estimation and data analysis.

Once all images had been segmented and written to a file, OpenCV is used to read the image data for use by Mediapipe. The Mediapipe pose estimation function then fetches the keypoint data of all images.

Initially, testing for this data was troublesome. The Mediapipe body pose estimation system tracks a total of 33 pose landmarks. Some of these landmarks are of no use to a golfer and therefore the data is not necessary. Due to this, only 15 of the 33 keypoints are used. these keypoints are shown in Fig 16. The location of these keypoints is shown in Fig 17.

```
keypoints = [11,12,13,14,15,16,23,24,25,26,27,28,29,30,31,32]
```

Figure 16 -necessary keypoints



0. nose
1. left_eye_inner
2. left_eye
3. left_eye_outer
4. right_eye_inner
5. right_eye
6. right_eye_outer
7. left_ear
8. right_ear
9. mouth_left
10. mouth_right
11. left_shoulder
12. right_shoulder
13. left_elbow
14. right_elbow
15. left_wrist
16. right_wrist
17. left_pinky
18. right_pinky
19. left_index
20. right_index
21. left_thumb
22. right_thumb
23. left_hip
24. right_hip
25. left_knee
26. right_knee
27. left_ankle
28. right_ankle
29. left_heel
30. right_heel
31. left_foot_index
32. right_foot_index

Figure 17 - keypoint locations

Mediapipe pose estimation returns three data axes, these being: the x-axis, y-axis, and z-axis. Due to this, factors such as distance from camera and footage resolution became pressing factors.

The normalised x and y axis are given 0 to 1, when the axes origin is the origin of the image axes-coordinate. This essentially means that the x and y axis are calculated relative to the footage scale, with keypoint data being a float value between 0 to 1 (MEDIAPIPE, 2023).

The normalised z axes are calculated relative to the z-origin of the wrist keypoint (MEDIAPIPE, 2023). This allows for approximate depth to be calculated. This opens a discussion into how player distance from the camera could affect the pose data.

To assess differences in footage resolution, an amateur swing moment was converted to 300x300 from 480 x 960. Fig 18 shows a plot of the original 480x960 resolution, Fig 19 shows a plot of the 300x300 resolution.

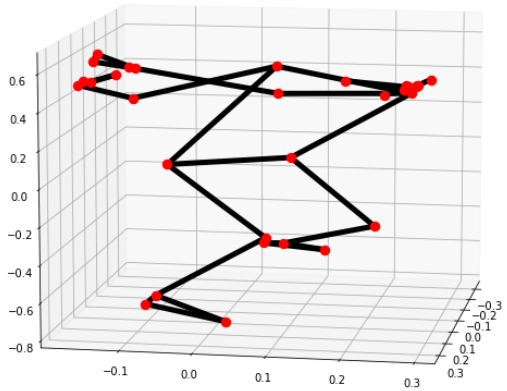


Figure 18 - 480x960 amateur resolution plot

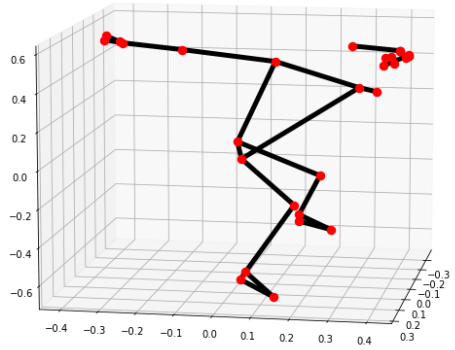


Figure 19 - 300x300 amateur resolution plot

From these pose plots, it is theorized that larger-resolution images may return more accurate pose data than low-resolution images.

Considering that the professional image data is of 160x160 resolution, the data quality of these images are in question. To test this, a professional swing image is tested 3 times, once at its original 160x160 resolution, once at 300x300, and once at 720 x 480 (common resolution of mobile phones) (DOROFTE, 2019). These resolutions are shown in Fig 20.

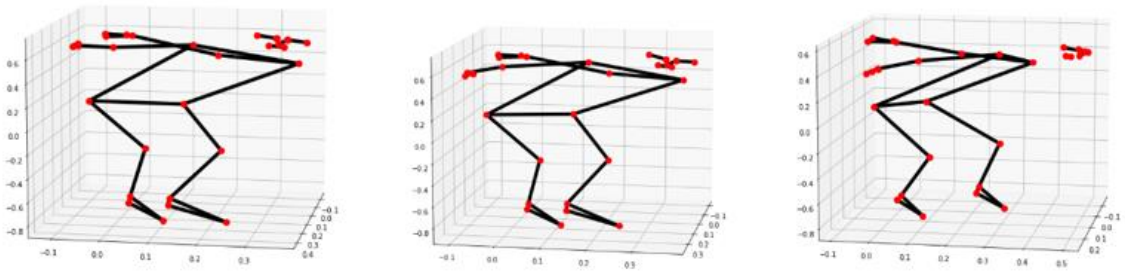


Figure 20 - (left) 160x160 resolution plot, (middle) 300x300 resolution plot, (right) 720 x 480 resolution plot.

Decidedly, the most erroneous pose resolution was 720x480, due to the difference in hand and shoulder keypoints.

This calls into question the ideal resolution of images when using Mediapipe pose estimation, Theoretically, all images should have the same x, y, and z axis due to the sportsperson being in the same position relative to the frame size, however, this is not the case.

At this point, the pose estimation method came into question, if pose data can vary to this extent, the average keypoints amongst professional golfers cannot return trustworthy comparison data. With a system so sensitive to change, factors such as frame or height of a golfer could affect the keypoint feedback.

Within the current system, if the average golfer data was male-centric, a female test subject would likely receive feedback on shoulder width or other physiological differences, despite this feedback being nonsensical. Considering these factors, two new systems are to be built and evaluated:

- Pose similarity detector – a system that will find the golfer with the most similar keypoints over all moments of swing.
- 2D pose estimation system – making use of the pose similarity detector, find whether testing on a 2D plane for pose similarity returns higher quality results than on a 3D plane.

With the use of a pose similarity detector, 2D keypoint data may return more meaningful data than 3D keypoint data as the problem of keypoint depth on 2D planes may not be such a large problem if the similarity detector can find a golfer in a similar location relative to the image frame.

Therefore, a 2D pose estimation system will be built with OpenPose, OpenPose is a real-time multi-person 2D pose estimation system that generates its Pose estimation through part affinity fields. OpenPose makes use of a 24-keypoint pose estimation system, as shown in Fig 21.

To determine the more accurate system, a comparison of the accuracy of the pose similarity detection system, when tested on both 2D and 3D pose estimation models, will be made through assessment of visual output.

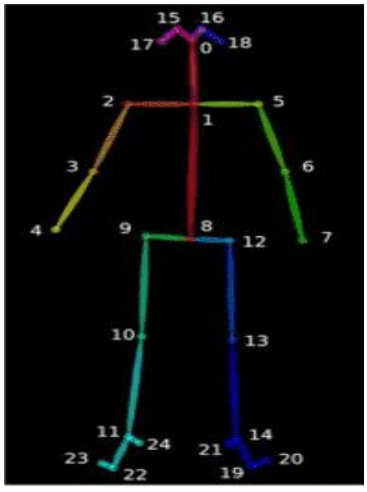


Figure 21 - OpenPose keypoints

OpenPose

A part affinity field is a representation of a set of flow fields that encode unstructured pairwise relationships between body parts of people. These flow fields are 2d vector fields that display the flow and direction of the connected body parts. New part affinity fields are generated for each separate body part connection. PAF calculates the chance that a pixel is a body part connection, where a higher PAF value means a higher confidence in detection (CAO, et al., 2021).

Part affinity fields are calculated as:

$$the\ set\ L = (L_1, L_2, \dots, L_c)\ has\ C\ vector\ fields, one\ pair\ limb, where\ L_c \in \mathbb{R}^{w \times h \times 2}, c\ \in \{1 \dots C\}$$

Images are input into the OpenPose system and OpenPose predicts both part confidence maps and part affinity fields, as shown in (b) and (c) in Fig 22. where confidence maps are denoted as S and PAFs are denoted as L. $w \times h$ is the size of the input image.

$S = (S1, S2, \dots, S_j)$, where j is the count of the number of body parts, $S_j \in \mathbb{R}^{w \times h}$, where $S_j\ is\ a\ w \times h\ matrix\ for\ j \in \{1 \dots j\}$.

$L = L_1, L_2, \dots, L_{c,}\ where\ C\ is\ the\ count\ of\ body\ part\ connections, L_c \in \mathbb{R}^{w \times h \times 2}$, where $L_c\ is\ a\ w \times h\ matrix\ for\ C \in \{1 \dots j\}$.

Part Confidence maps are the chance that body part areas are in the image, where there are J maps per body part. Sj returns the confidence map for each jth body part.

PAFs encode the relationships of the body parts. Where there are C vector fields per body part connection. Each vector field stores 2D vectors which show the relationships between each body part connection. L_c returns the part affinity field for each Cth body part connection.

The Part affinity fields and part confidence maps are then parsed by greedy inference to output the 2d keypoints for all people identified in the input image (CAO, et al., 2021).

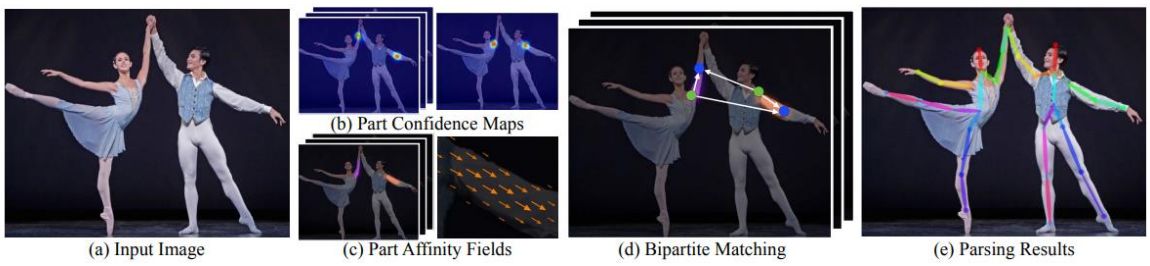


Figure 22 - OpenPose pose detection pipeline.

In comparison, Mediapipe uses BlazePose, which is a lightweight CNN architecture that makes use of heatmaps, offset, and regression to predict the location of the 33 keypoints (BAZAREVSKY, et al., 2020). To detect the person, a detector-tracker system is used due to its excellent performance of landmark tracking shown by (KARTYNNIK, et al., 2019). The tracker predicts key point coordinates of the object in the current frame, as well as a region of interest for the current frame. The detector-tracker system is shown in Fig 23:

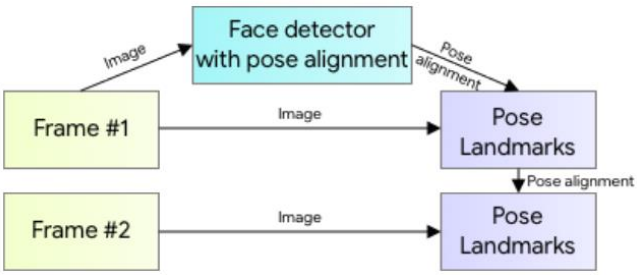


Figure 23 - BlazePose detector-tracker system

Once completed, the aforementioned combination of heatmaps, offset, and regression are used to remove the output layers from the model before running inference. To achieve a balance between the high- and low-level features, skip connections are utilized within all stages of the network. By limiting the number of scales, translations, and angles during data preparation, the architecture is made faster by decreasing the computational load. To increase pose quality, the pose object (person) is aligned through the point between their hips. The image is converted to fit a square bounding box around the body. Through This, the tracker can maintain respectable tracking rates, even in cases of significant occlusions (BAZAREVSKY, et al., 2020).

As shown in Fig 23, the pose data is unstable when tested over multiple resolutions. Fig 24 shows the OpenPose pose skeleton tested on the same 3 images of varying resolution sizes.



Figure 24 - (left) 160x160 resolution plot, (middle) 300x300 resolution plot, (right) 720 x 480 resolution plot.

From Fig 24, it is inferred that OpenPose pose estimation returns more consistent and accurate keypoint data between resolutions. However, the quality of data is still in question. As OpenPose uses a 2D pose estimation system, factors such as keypoint depth are a concern. OpenPose does not allow for the abstraction of pose data in the same format as shown in Fig 20, so the accuracy of OpenPose pose visualisation, shown in Fig 24, cannot be compared to the same standard.

Both systems have been shown to have positives and negatives, OpenPose pose estimation is more accurate over differing resolutions and returns a more readable visual output. Mediapipe pose estimation is less accurate and readable, however, is tested on a 3-dimensional plane which accounts for depth. The importance of keypoint depth cannot be understated as without this extra dimension, accurate spatial positioning cannot be assured.

Pose Comparison

OpenPose pose data can only be written to a JSON file and therefore Mediapipe keypoint data is also written to JSON for ease of use.

To Compare keypoint data, the Euclidean distance between the two points in a line segment must be found, where the two points are the amateur keypoint average and the professional keypoint average.

To calculate one-dimensional Euclidean distance, the distance between two keypoints on a number line is the modulus of the difference of their coordinates, defined as the absolute difference. Two points are defined as p and q, on the number line, the difference between them is found through the formula (SMITH, 2013):

$$d(p, q) = |p - q|$$

$$Where\ p\ =\ \text{amateur keypoint},\ and\ q\ =\ \text{professional keypoint}$$

The Euclidean distance between the amateur and the professional keypoints are calculated to find professional golfer with the lowest average keypoint distance from the amateur.

Through this technique, both pose systems return the golfer with the most similar pose data. Interestingly, despite the test subject being a 5ft 6in male, the Mediapipe data returned golfer Lexi Thompson, a 6ft tall woman as the most similar golfer. OpenPose returned golfer Nick Watney, a 6ft 2in man, as the most similar golfer. This could be both systems accounting for the difference in player depth between amateur and professional data, where taller golfers are returned as the amateur player is closer to the camera than the professionals, therefore filling the image an equal amount despite being of smaller size.

It should be noted that OpenPose does not have pre-built pose abstraction capabilities, and a separate method must be used.

While all keypoints are detected, some do not return in the plot. Appendix A: fig(s) 33-34 shows Mediapipe pose comparison, Appendix A: fig 35 shows OpenPose Pose comparison. The amateur pose is shown on the left, professional pose on the right.

OpenPose cannot visualise keypoints from input, meaning the amateur average cannot be truly represented by a single pose. so, the average pose is found by overlaying multiple images on top of one another.

In the Opinion of the researcher, Mediapipe Pose estimation will be the better choice for Swing analysis based on visual output. The 2-dimensional data within the OpenPose system is not quality enough to discount Mediapipe's use of keypoint depth.

Considering that missing limbs are not ideal for data analysis (as shown in Figs 33 and 34), rather than using the single top golfer, the top 3 closest golfers will be used for pose comparison, the plot comparison is shown in Appendix A: fig 36, 37.

From these visualisations of the data, it is decided that the technique of using the top 3 closest poses is the best way to find comparison data due to the potential for erroneous data found by using just 1 professional. The plots in Fig (s) 35 and 37 show that the amateur and professional swings are close enough so that they will not return erroneous data, while also different enough so that changes can be recommended to the amateur swing.

To return feedback on the golf swing, keypoints will be grouped per location on the body, where a group of keypoints make up a joint, this is so that a full area of improvement can be identified rather than a single keypoint. Considering the dynamicity of this data, the open AI library was used to prompt a natural language processing model that could read the data and return dynamic feedback based on the data. The GPT-3 Model "text-DaVinci-003" was used due to the relatively low running costs of this model and high-quality text inference abilities.

The GPT-3 model is an autoregressive language model with 175 billion parameters, over 10x more than other language models available at the time of release. GPT-3 achieves quality performance on various NLP problems, such as question-answering and cloze tasks (BROWN, et al., 2020). The GPT architecture makes use of a transformer architecture to return realistic responses to text-based prompts (VASWANI, et al., 2017). The transformer architecture is based purely on attention mechanisms, such as scaled dot-product attention, in which the input consists of queries and keys of dimension d_k and values of dimension d_v. The dot products of the queries with all keys are computed. Each query is divided by $\sqrt{d_k}$, and a SoftMax function is applied to obtain the weights of the values (VASWANI, et al., 2017). Scaled Dot-Attention will compute the correlation between the query, value, and key representation of the input tokens. The scaled dot-product Attention mechanism operates within a layer known as the Multi-Head Self-Attention" layer. This layer uses multiple Attention heads, these heads have their own learnable parameters which allow the model to focus on unique features of the input at the same time. This allows for different relationships between words to be concentrated on simultaneously.

Through this model, active and dynamic interpretation of the swing data can be made.

Chapter 5: system evaluation.

To test this system, the inferences made on the data by the GPT-3 Model were relayed to the amateur golfer.

Two factors will be used to dictate whether the research has proved a success or a failure, these being: golfer sentiment and drive distance.

The system may return meaningful feedback without an improvement in golf drive, this could be due to numerous factors such as the golfer's physical condition, as well as wind strengths and ball/club quality. As well as this, the system could return meaningless data, but the drive distance could improve due to the same factors.

The GPT3 system feedback is shown in Figs 25 and 26.

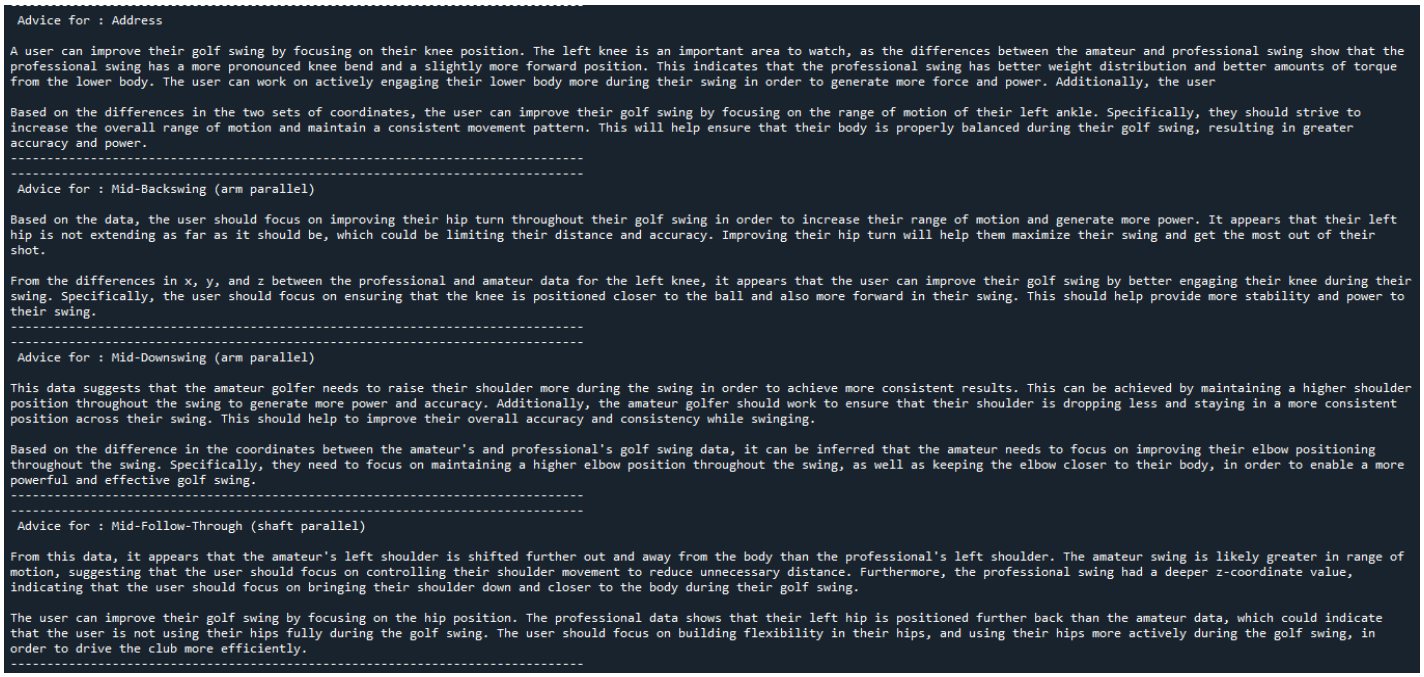


Figure 25 -Feedback 1

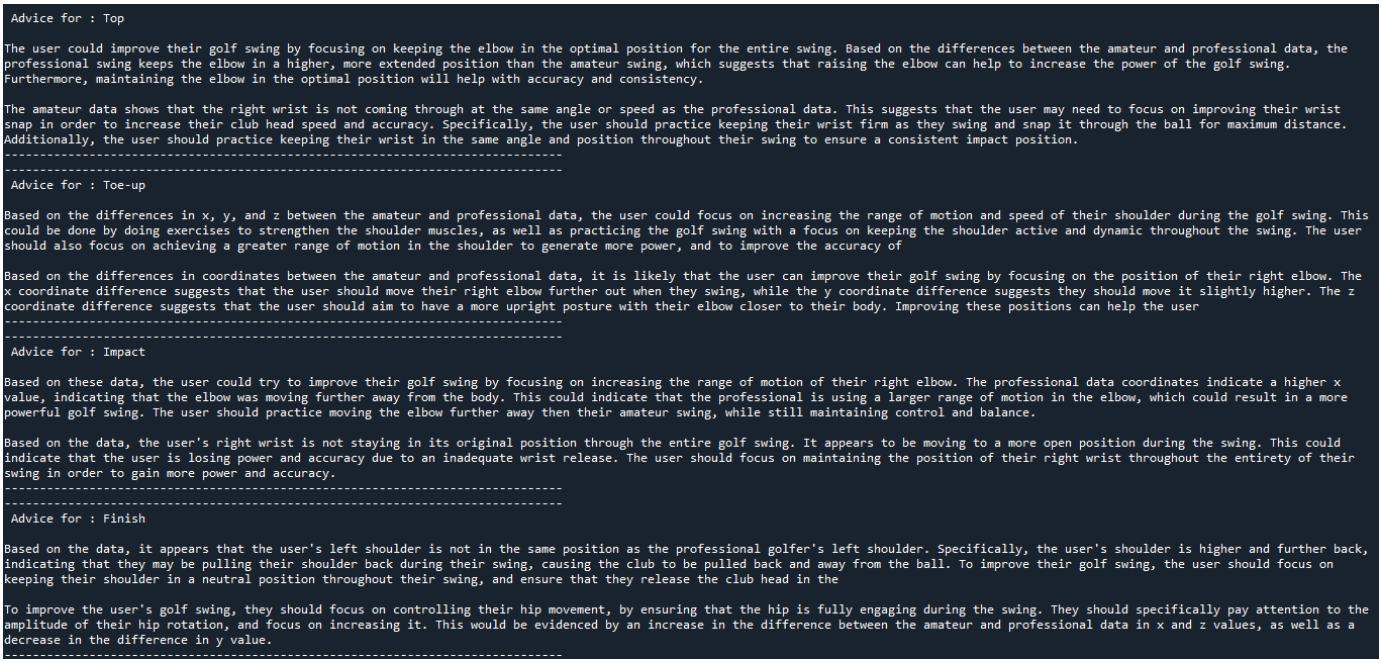


Figure 26 -Feedback 2

This amount of data can be overwhelming for an amateur sportsperson; therefore, the sportsperson chose three sections of their swing that they feel could be improved. The sections chosen were the top, Toe-up, and Impact.

As can be seen in Fig 25, the feedback places a strong emphasis on the right arm. Specifically, the wrist and elbow, are mentioned in both the top and follow-through. The Feedback system finds that the sportsperson is not bending either their wrist or elbow to the recommended level, this can be shown in Fig 25, wherein both the Top and Impact, the angle at which either the wrist or elbow sits could be improved. Fig 26 shows a comparison of the amateur golfer's top compared to a professional player's top before feedback was administered. As identified, the amateur player is not bending either the wrist or elbow enough to reach an ideal moment of swing.



Figure 27 - pre-advice comparison

In the initial amateur footage, the average range of the golf drive was found to be between the 150-175 yards range. Once the sportsperson was told to bend their wrists at the Top and keep the elbow closer to the body on the impact the sportsperson was able to drive between a range of 160-200+ yards, with the furthest distance recorded at an estimated 225 yards.

As well as this, the sportsperson acknowledged that they were unaware of this drawback to their game and that changing their pose accordingly improved the quality of the swing. The sentiment of this sportsperson was overwhelmingly positive due to a clear improvement made in the quality of their golf swing. Fig 28 shows a comparison of the top once the sportsperson had begun employing the feedback to their swing.



Figure 28 - post advice comparison

As shown, the sportsperson brings the club further over their head than before, resulting in a higher quality swing.

Chapter 6: conclusion/ critical evaluation

Project summary

The system attempts to address the problem of improving amateur golf performance, through pose estimation techniques and personalized feedback for improvement. Typically, golf analysis would come from an expert such as a coach, but the opinions provided by these experts can be subjective based on the expert's interpretation and opinions. This system attempts to use ML techniques to efficiently automate a golf swing analysis to provide unbiased feedback through a comparison to a professional sportsperson's swing data.

This system aimed to improve the drive distance of an amateur golfer by comparing their swing to a professional at 8 different key moments of a golf swing. This was fulfilled by informing an amateur golf player of the feedback the system had provided, and assessing whether there was an improvement in the golf swing.

To access the 8 different key moments of a golf swing, an image segmentation system named "golfDB" was used to sequence and segment both amateur and professional footage into images for comparison of these respective moments. These moments were compared by applying a pose estimation library known as "Mediapipe" to each image. Mediapipe analyses the body of the sportsperson in question and maps their body with "keypoints", which are important locations on the body, such as the knee, or elbow. Each keypoint is given an x, y, and z value for their location in a 3-dimensional space. For each moment of swing image, the pose estimation library is applied, and an overall average for each keypoint is calculated for each amateur and professional at each moment of swing, this system assumes that only 1 amateur is being analysed at a time. Once this keypoint data is calculated it is written to a JSON file for data readability. A pose comparison system identifies the top 3 professionals whose keypoints are on average closest to the amateur keypoints average, and the averages of these 3 professionals' keypoints are found and returned to act as the professional pose that the amateur pose is compared against. Finally, the keypoint data of both the amateur and the professional are inserted into a GPT3 language model that returns inference on the data and provides feedback on how the user can improve their swing at each key moment of swing. The success of the system was based upon two factors: amateur golfer sentiment and drive distance.

Through these methods, the amateur golfer shown an increase in swing distance of 15% - 30%. As well as this the user remarked that the system was correct in the areas that needed improvement and that they were previously unaware of the problem areas found by the system. This level of improvement cannot be promised for other golfers due to several factors, such as willingness to change technique as well as room for improvement.

This system will not work with golfers who are such amateurs that their swing is too different from the professional players to be detected by the swing segmentation system. The swing segmentation system was trained on professional players and therefore the swing needs to be similar to an extent. This system only accounts for keypoint location at the moment of swing, therefore factors such as swing speed and club location are not accounted for, and yet they are especially important aspects of the golf swing. Club positioning is arguably just as important as body positioning, The club location can be assumed based on the location of the wrist keypoints, however, it cannot be guaranteed and therefore is a large limitation. As well as this, the current system only accounts for the use of the "Driver" club. This isolates large parts of the game of golf from analysis and for a system to be able to state that it can improve golf performance, it should ideally apply to all areas of golf. Users of non-standard body posture, such as disabled individuals could likely not use this system as their body positioning may be so non-standard that it could not be reasonably compared to that of an able-bodied person.

Critical evaluation

Requirements.

The functional and non-functional requirements were all met to an extent, The methodology for the meeting of certain requirements may have been altered due to changes in the system. This is primarily shown in requirements based on keypoint data and professional comparison. As shown in the "Pose Comparison" section of Chapter 4, changes to how the swings are compared resulted in more ideal comparison data. As well as this, the initial requirement of the comparison of individual keypoints was scrapped in place of a comparison of multiple keypoints that make up a limb. This is to give a larger overview of what can be improved rather than a more specific piece of feedback.

The requirement of rejection of low-quality footage was not employed in the building and testing of this system due to time constraints. Considering that the amateur swing footage was recorded away from the computer the system was built on, having to go back to the driving range the footage was recorded at was not a possibility due to the amateur golfer's availability. Initially, the system would reject any footage in which any moment of swing has an accuracy rate of less than 0.125. however, only 12 videos were taken of the amateur's pre-feedback swing. Due to the unpredictability of this footage when presented to the segmentation system, the accuracy requirement was removed. If this system were ever deployed to a real-world scenario, such as an app, the accuracy requirement would be re-introduced as access to data would be under less constraint.

The non-functional requirements state that a level of scalability should be possible, however, the system is already scaled to a decent level, being able to process over 220 professional swings in under 2 minutes. The largest constraint in terms of scalability would be the GPT-based feedback system. An individual can only use $120 worth of tokens with this model, at $0.02 per 1000 tokens, the system would eventually hit its limit if it had enough users. So far, the only way to bypass this limit is to apply for use as a business, but this would require a business association which the developer does not have.

Using tiktoken, a library for counting the number of tokens used in a piece of text, it is calculated that roughly 1500 tokens are used in a full piece of feedback. Thus costing $0.03, meaning that roughly 4000 users could use the system before reaching the prompt limit. An example, as well as the token amount, is shown in Figs 29 and 30.

```python
import tiktoken

def num_tokens_from_string(string: str, encoding_name: str) -> int:
    """Returns the number of tokens in a text string."""
    encoding = tiktoken.get_encoding(encoding_name)
    num_tokens = len(encoding.encode(string))
    return num_tokens

text = "Comparing the amateur and professional swing, it appears that the user is not ge

num_tokens = num_tokens_from_string(text, "p50k_base")
print(num_tokens)
```

Figure 29 - tiktoken code

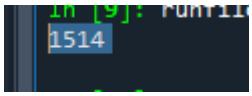


Figure 30 - token amount

Design

The Python programming language was chosen due to its strengths in lightweight development environments and data analytics. As well as this, there was little choice as GolfDB was developed in Python, and the decision to keep this system built within one programming language allowed for ease of development.

When considering the outline for this system, the golfDB segmentation system was always considered necessary for image segmentation due to it being pre-trained and could be employed immediately in the system. Other proposed segmentation methods that rely on motion entropy analysis and piece-wise linear approximation could have been considered, however, As the golfDB could be instantly deployed, the time used on researching such methods would likely not yield worthy results.

The Mediapipe Pose estimation model was considered to be the best option for pose estimation as it could easily gather the keypoint data of the pose for efficient data returns. However, the use of motion synchronization and motion discrepancy has been used to develop other swing analysis tools (LIAO, et al., 2022). This method was not used in this project due to a lack of useable models. Mediapipe offers a wide range of pose estimation models for various use cases, so developing a custom model would not be worth the time and effort, especially considering that there is no guarantee it would be any more accurate.

Initially, the plan for how feedback would be generated was unclear, the primary idea was that the keypoint with the largest Euclidean distance from the professional would be returned as the "area of improvement". However, this limits the quality of the feedback that could be returned. With no context as to how the keypoint was different from the professional average, only very general feedback could be returned. With the use of the GPT-3 language model, dynamic inference can be made on the data, as if it was interpreted by a person. The quality of the feedback is proved to be extremely high quality, as shown in Figs 25 and 26.

implementation

The Python programming language was chosen due to its strengths in lightweight development environments and data analytics. As well as this, there was little choice as GolfDB was developed in Python, and the decision to keep this system built within one programming language allowed for ease of development. The system could have been built in C++ but there were very few upsides in doing so as libraries such as OpenCV and Mediapipe are less accessible in C++ than they are in Python. There are implementations for both of these libraries in C++, but due to the lightweight nature of the Python programming language, it can return data faster and more efficiently than C++ can (MCKINNEY, 2012).

Accuracy of swing segmentation

The GolfDB implementation of the swingnet baseline model was trained on high-quality video data of a frame size 160x160. The frame size of the test footage was 720x480 and therefore could have led to a drop in accuracy rates in the swing segmentation system as due to the difference in aspect ratio. This should have been considered when evaluating the performance of this system. if the system were to be deployed to a consumer-level product, footage from mobile phones that do not record in 160x160 would likely be used. Differences in resolution should have been discussed and assessed further. The developers of the swing segmentation system tested a maximum of 224x224 resolution, finding that 160x160 was more cost-effective (MCNALLY, et al., 2019). A significantly larger difference in resolution, as used in the amateur test footage could result in significant drops in accuracy.

When segmenting images, a drop in accuracy rates between the First and last moments of the swing is shown in Figs 4 and 5. These are the address and finish moments. The developers of GolfDB suggest that this is caused by the subjective labelling of these events during model training (MCNALLY, et al., 2019).

Due to the GolfDB system being trained on professional swings, it may struggle when segmenting amateur swings. The swings performed by an amateur may be so different from a professional that the system may not be able to detect the moment of swing due to the significant difference between the two. Therefore, it is concluded that the amateur golfer must have some level of experience in golf and be able to perform relatively uniform golf swings. Below are the accuracy rates for each moment

of swing. Fig 31 shows the average accuracy rates per moment of swing for professionals, and Fig 32 for amateurs.

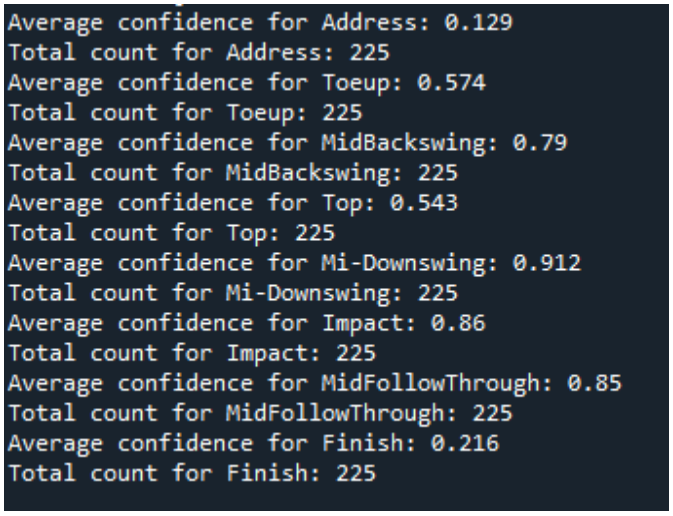


Figure 31 - professional accuracy rates

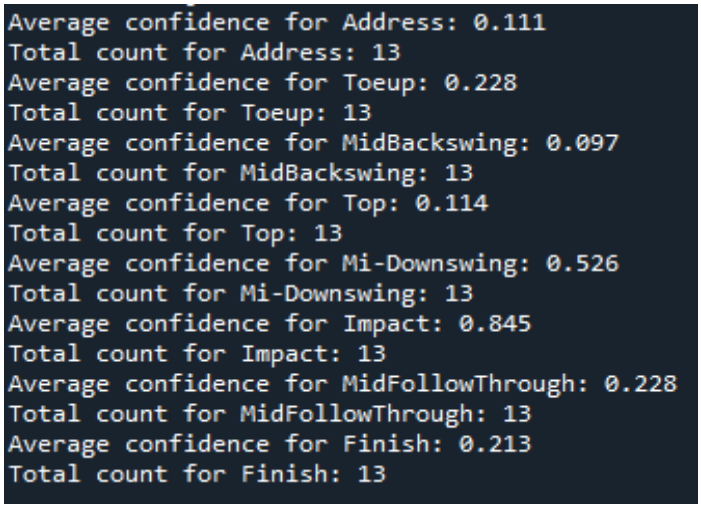


Figure 32 - amateur accuracy rates

In conclusion, the system performed to a satisfactory standard when detecting moments of swing for professional golfers, however, some further improvements could be made to accommodate amateur golfers.

Quality of pose estimation and Pose comparison.

Pose estimation.

Pose estimation is a crucial feature of the golf swing analysis system. The use of 3D pose estimation over 2D pose estimation has been employed due to the z-axis data 3D pose estimation can capture, this allows for depth of pose to be captured. However, 3D pose estimation systems such as Mediapipe can struggle with challenges such as accurate readings as shown in Fig 20. The figure shows pose plots can change depending on image resolution, indicating the instability of the Mediapipe pose estimation system. The similarity of resolution must be stressed due to its unforeseen effect on both the segmentation and pose estimation systems. Even though Mediapipe was chosen over OpenPose, factors such as accuracy, precision, and consistency should have been considered further. As stated in Chapter 4, OpenPose returns better accuracy and precision amongst different resolutions when comparing Fig 20 to Fig 24. Thus, it is important to discuss whether a 2D pose could be analysed deeper than a less accurate 3d pose, especially considering that this data will be inferred by the GPT3 model.

Despite this, the feedback system still returned meaningful inferences on the amateur golf swing, so this problem may not be as large a problem as it appears to be.

The reasons why Mediapipe was chosen over OpenPose have already been discussed at length, however, manual analysis of visual output may not be the most ideal way of deciding the quality of these systems, and factors such as accuracy, precision, and consistency should have been considered further. As stated in Chapter 4, OpenPose returns better accuracy and precision, amongst different resolutions when comparing Fig 20 to Fig 24. As this data will be inferred by the GPT3 model, it is necessary to consider whether a more accurate 2D pose could be analysed deeper than a less accurate 3D pose.

Pose comparison.

Regarding pose comparison, the end decision to use the top 3 closest poses cannot be faulted, as shown in fig 33 and 34, comparison of just one professional resulted in missing keypoints as only one pose was being assessed, and the comparison of all professional players would have resulted in lower quality comparison data. A discussion into whether the use of the top 5 or 10 poses would have been ideal could be opened, however, this could only result in more dissimilar poses being used for comparison.

It could be argued that using the average pose would be better, as using the most similar poses creates less space for criticism, if the professionals are doing anything incorrectly, it will not be detected. However, this is considered a necessary course of action. It is unlikely that an amateur sportsperson will change their entire form based on this system, rather small pieces of advice are better, which is what is provided by using the top 3 closest golfers, as using the full professional AVG would result in an overwhelming number of changes recommended due to the vast differences.

In summary, while pose estimation is a key element of the system, potential issues, and unanswered questions lead to many factors to be considered as to how the most ideal data can be returned. The use of 3D pose estimation and the pose comparison method are crucial factors that should be critically evaluated further.

Quality of feedback

The use of the GPT3 language model is an innovative approach for dynamic feedback, however, it could be argued that it is not the best fit for this system. Considering the previously discussed cost of use, as well as its lack of proven testing on returning inference on systems such as this, problems could arise. As shown in Fig 25-26, some feedback returned by GPT3 discusses the technical aspects of the differences in the golf swing, mentioning the x, y, and z-axis, however many users may not know what this means and therefore cannot understand the feedback. As well as this the quality of feedback could be questioned, as seen in the Toe-up section of Fig 26, the system recommends that the user could perform exercises to improve shoulder strength. While this is not poor feedback, it isn't exactly ideal for our means, as clear feedback on the technique is more ideal.

Despite these factors, the GPT3 model provides feedback of a quality that would not be possible without such a high-quality Deep learning NLP system.

Future work/ directions

In the future, application into the wider game of golf may be considered, At the moment, only the drive swing is accounted for, however, this excludes a sizeable portion of the game of golf, with further application, a full golf swing improvement system could be developed for performance improvement.

Club analysis could be built into a pose estimation system through deep learning techniques. The lack of consideration for club speed and location could mean that the system is returning lower quality data than what is possible. The inclusion of these features would improve the system to a higher level than what current pose estimation can give.

The development of more motion classification systems such as golfDB is necessary to build the field of data-driven sports development. Due to the relatively uniform motion of a golf swing, the labelling of these motions would not be troublesome. However, sports in which there is not one uniform motion to perform a task, such as a free kick in football, would take much more time to produce image segmentation systems, and alternative methods for automated performance improvement could be considered.

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Appendices

Appendix A: amateur and professional comparison plots

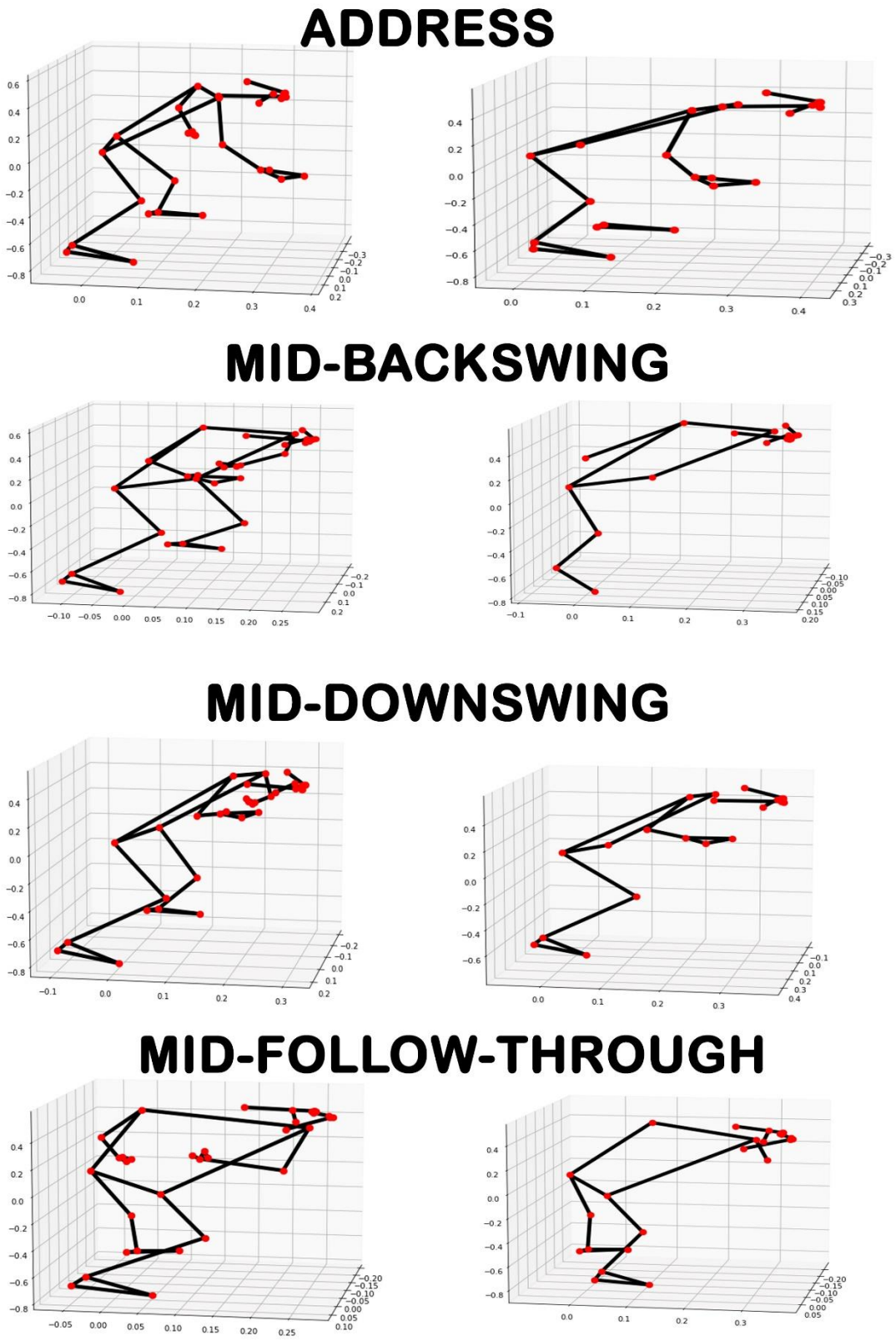


Figure 33 - Mediapipe pose comparison 1.

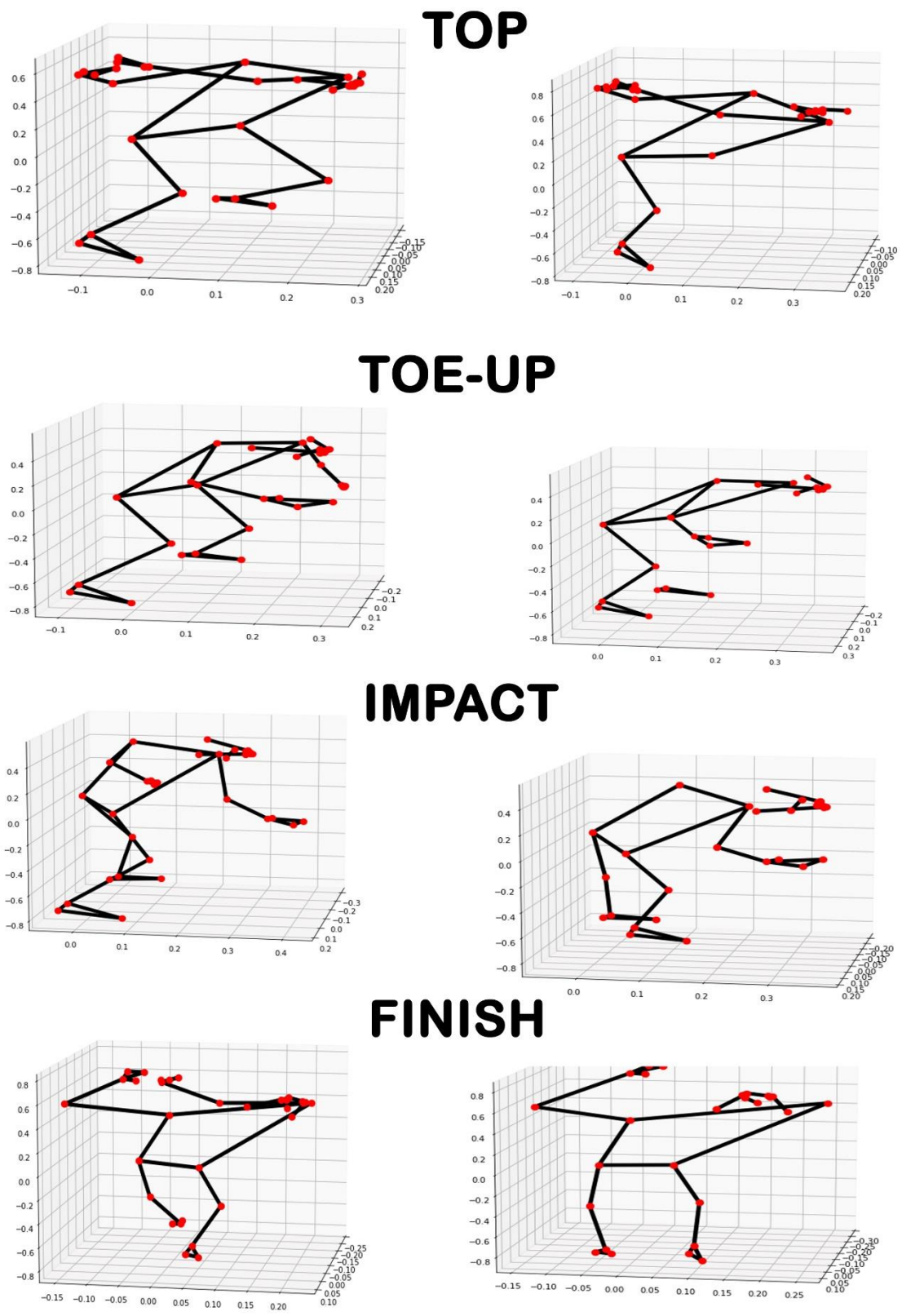


Figure 34 - Mediapipe pose comparison 2.

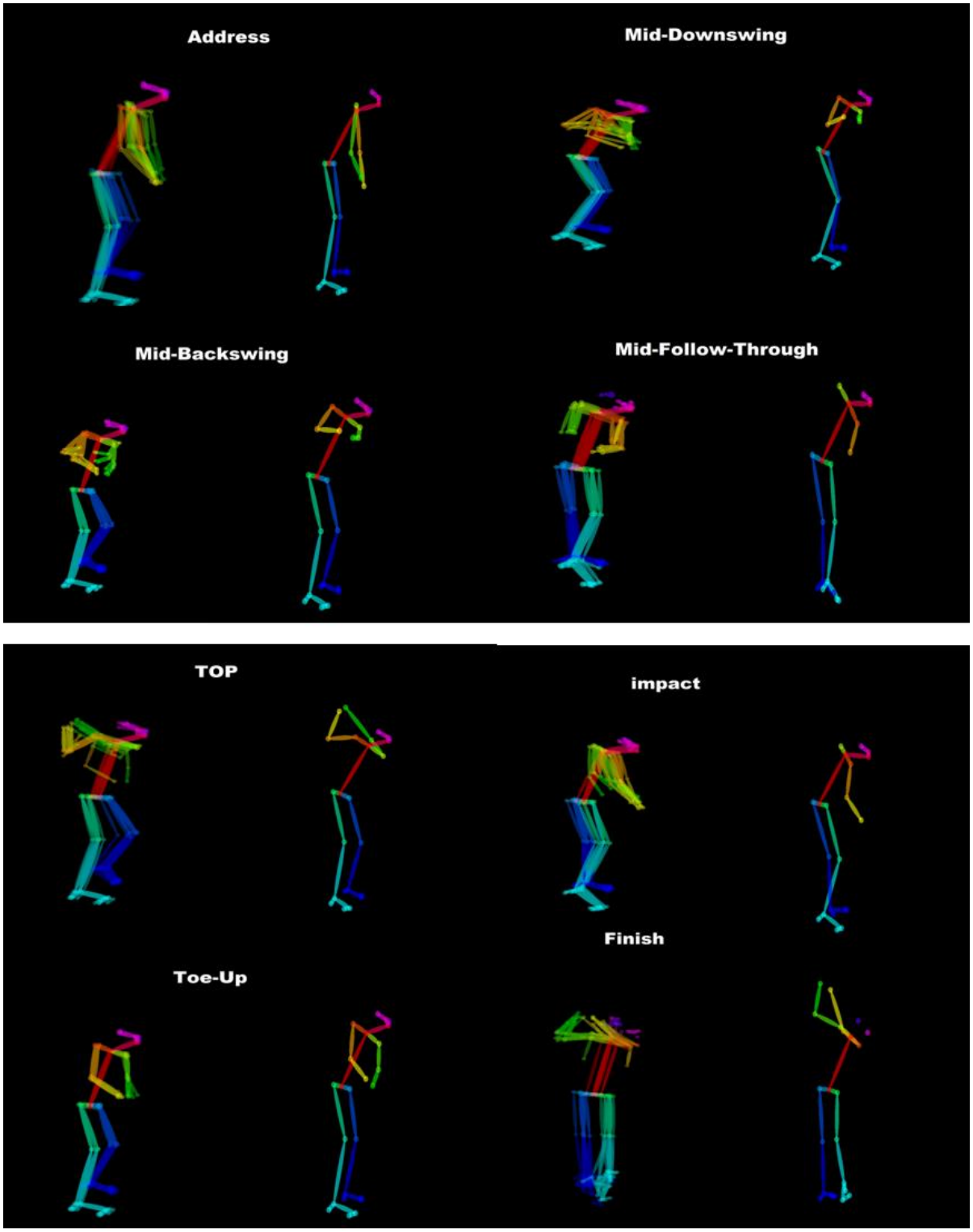


Figure 35 - OpenPose pose visualisation.

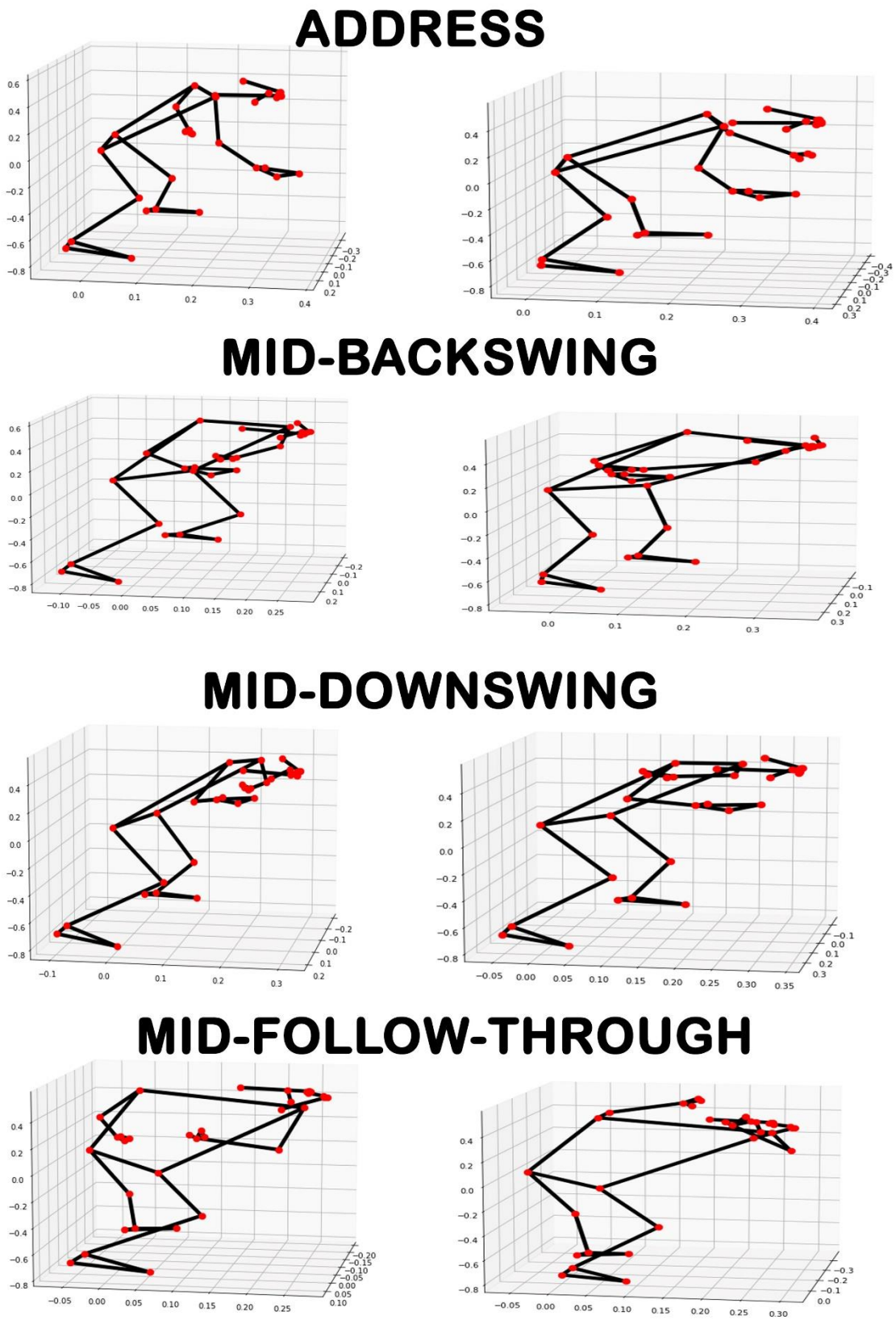


Figure 36 - Mediapipe combined professional comparison 1

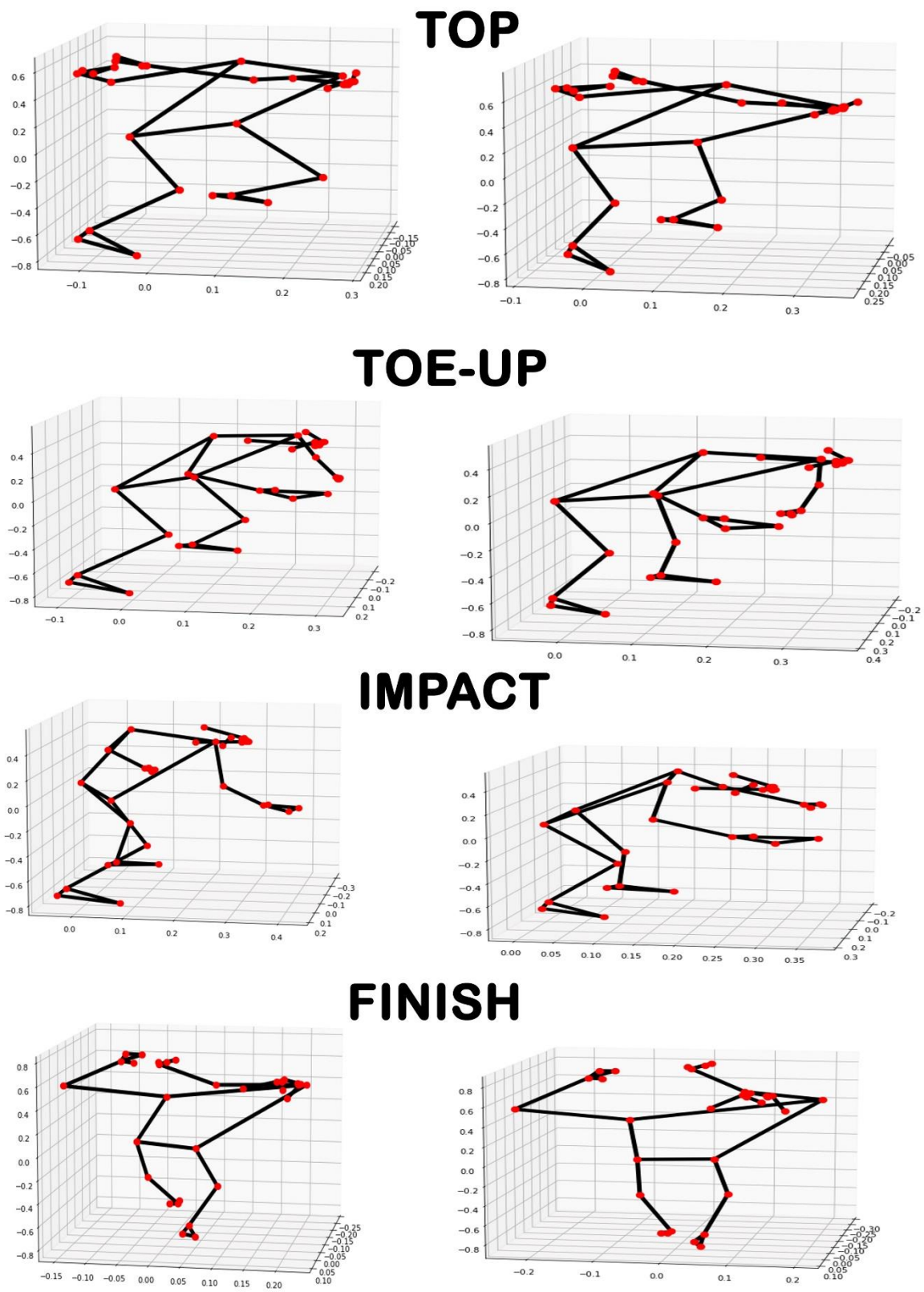


Figure 37 - Mediapipe combined professional comparison 2.

Appendix B: user manual

Library requirements:

- openAI
- math
- json
- pytorch
- numpy
- OpenCv
- Mediapipe
- OS
- math

To begin, follow the steps shown at: https://github.com/wmcnally/golfdb to acquire the necessary pre-requisites for the golf sequencing/ image segmentation system.

All golfDB models other than the one used have been removed, if wanted, generate using “train.py”

The full video repository has also been removed, only the necessary videos remain, if access to the full repository is needed, download at :

https://drive.google.com/file/d/1uBwRxFxW04EqG87VCoX3l6vXeV5T5JYJ/view

and place in “data” file

If any problems arise within the image segmentation system, clone the repository shown above and insert both “ProfessionalSegmentation” and “AmateurSegmentation” into the golfDB directory.

Within the GolfDB directory,

To segment the professional golf footage, run “ProfessionalSegmentation.py”

To segment the amateur footage, run “amateur segmentation.py”

The segmented images are stored in their respective files within “ImageData.py”

Go to the Pose_estimation folder stored in the root directory

Run both “AmateurKeypointToJson.py” and “ProfessionalKeypointToJson.py”, this will store the keypoint data in respective Json Files.

Run “JsonComparison.py” to find the Top 3 closest golfers in terms of Euclidean distance.

Run “JsonAvg.py” to find the average keypoint data of these 3 golfers.

Go to the root directory and run “feedback.py” to return feedback data.

Please note that the openAI api key provided within Feedback.py is my own personal API key. I have no problems with the supervisors using this key as much as they like, however Please do not share this API to anyone that does not need access to it.

If there is any problem with the API key, please let me know so a new one can be provided.